

A framework to incorporate lived experiences with aided communication systems to enhance the expression of socio-emotional concepts

Mathieu DePaul, Michael Cabrera, Kyra Dang, Amitya Thimaiah, Nathan Huey

Advisors: Dr. Alyssa Kubota, John Kim

School of Engineering & Department of Speech Language and Hearing Sciences



SAN FRANCISCO
STATE UNIVERSITY

BACKGROUND

- 8% of children in the US have disabilities that affect their ability to use speech to communicate [1, 2].
- Many use augmentative and alternative communication (AAC) systems to augment or replace oral speech. These systems can increase participation, support comprehension, and decrease frustration among users [3].
- However, they typically utilize generic symbols (see Fig. 1) which limit a user's ability to convey abstract socio-emotional concepts (e.g., feelings, states of being) [4, 5].
- Funds of knowledge (FOK), or a person's cultural and lived experiences from their homes and communities, may help facilitate the learning and expression of these socio-emotional concepts.

INCORPORATING AIDED COMMUNICATION SYSTEMS WITH FUNDS OF KNOWLEDGE

We propose leveraging FOK as a personalized and intuitive means of expressing socio-emotional concepts through AAC systems. We will continue to refine and expand on these key concepts in collaboration with families that use these systems.

Visual Scene Displays (VSDs) to capture lived experiences

VSDs (see Fig. 2) are an alternative to the traditional grid displays and serve as a natural opportunity to capture familiar images from a user's life. By embedding users' lived experiences into VSDs, communication may be developed in a context that is both familiar and of interest to the child. By rooting VSDs in familiar scenes from the child's own world, VSDs expand beyond basic wants and needs, opening opportunities for emotional expression and connection.

Strengths-based over deficit-based approach

A strengths-based approach shifts the focus on students' abilities, rather than their perceived deficiencies, especially when supporting communication and language for nonspeaking children. By highlighting these strengths, this approach boosts confidence, increases engagement with the proposed framework, and encourages language development.

FUTURE WORK AND EXPECTED OUTCOMES

1. To **identify prevalent representations of socio-emotional concepts**, we will conduct surveys and interviews to understand how aspects of daily life correspond with a user's lived experiences.
2. We will integrate our findings on a social robot to **enhance existing AAC systems and supplemental devices**.
3. Finally, we will **evaluate our system with nonspeaking children** using mixed methods, including engagement measures and system efficacy.

We anticipate that this work has the potential to significantly increase understanding of how nonspeaking children understand and express socio-emotional concepts.

RESEARCH QUESTIONS

This project aims to incorporate FOK in AAC systems to enhance the learning and expression of socio-emotional concepts.

RQ1: What are salient features of common representations of abstract socio-emotional concepts, drawn from a person's FOK?

RQ2: How we design AAC systems to effectively incorporate these features to enhance the expression of these concepts?

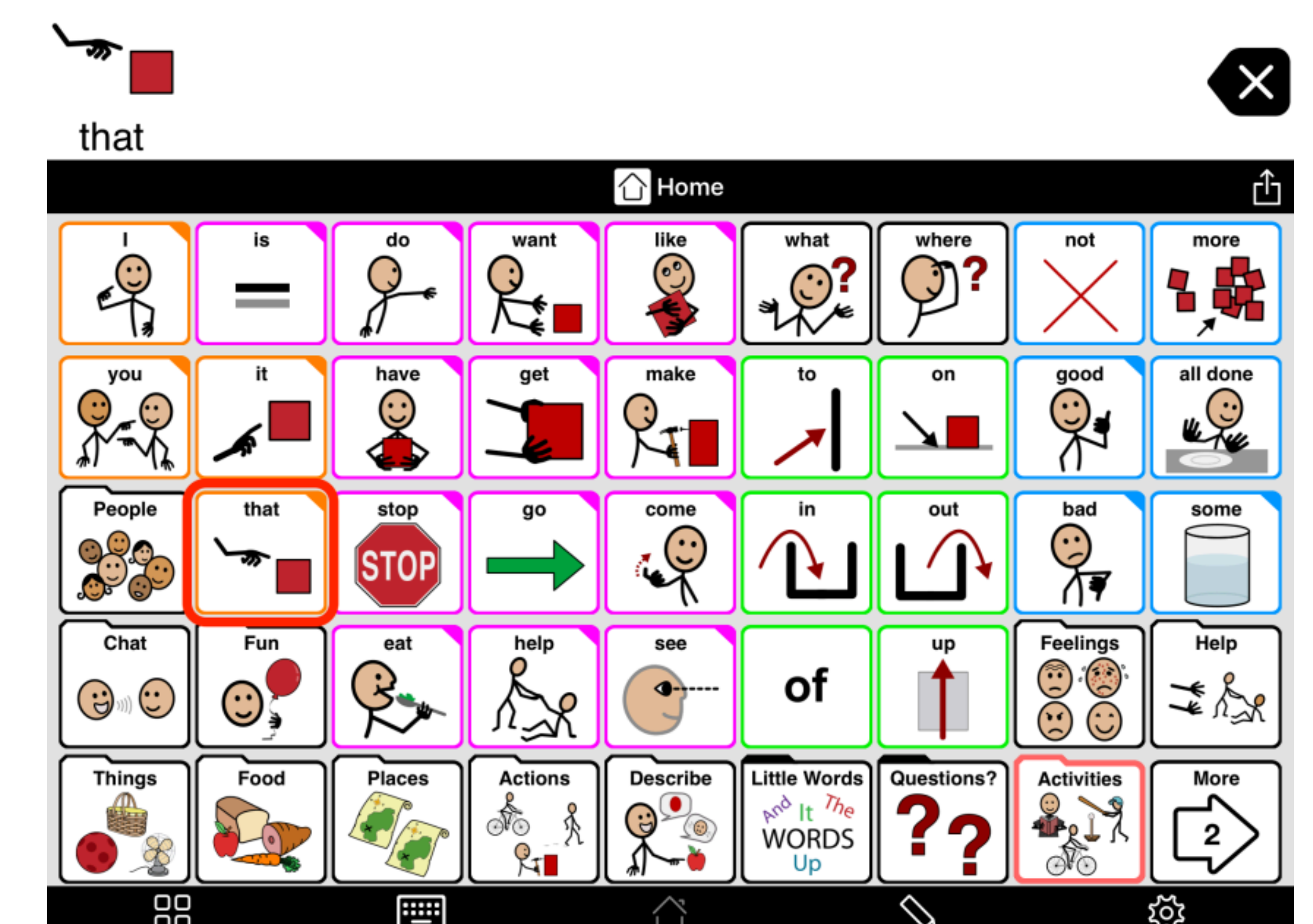


Fig 1: A traditional grid-based AAC system

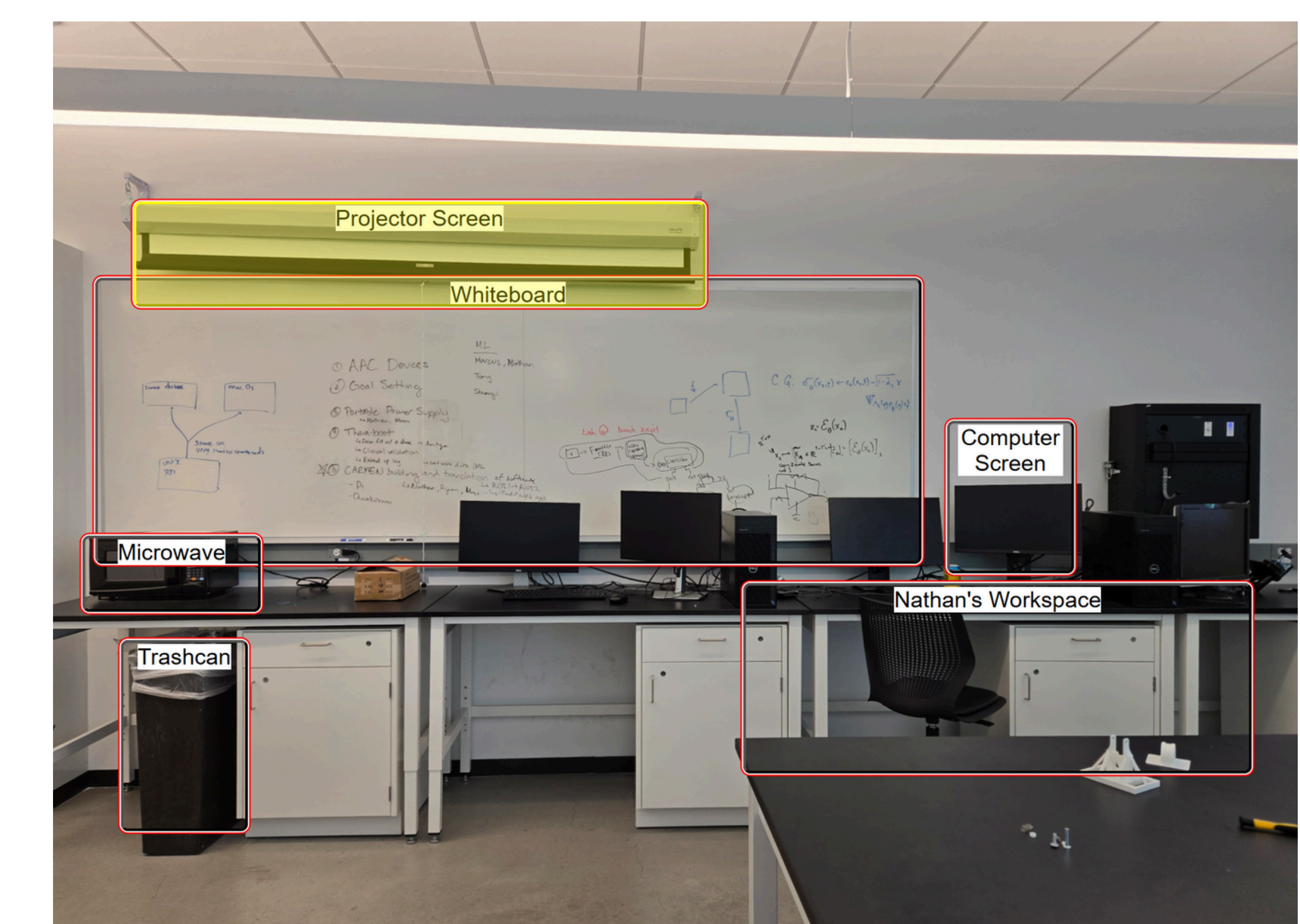


Fig 2: A visual scene display AAC system